

Adopted Levels

$Q(\beta^-) = -21910$  syst;  $S(n) = 17770$  syst;  $S(p) = 1.03 \times 10^3$  11;  $Q(\alpha) = -8.94 \times 10^3$  11 2021Wa16  
 $S(p), Q(\alpha)$ : Deduced by evaluators using mass excesses of 4777 105 for  $^{35}\text{Ca}$  measured by 2023La09,  $-1487$  17 for  $^{34}\text{K}$  measured by 2024Dr01, and 11290 16 for  $^{31}\text{Ar}$  measured by 2024Yu13. Values from 2021Wa16:  $S(p) = 880$  280 (syst),  $Q(\alpha) = -8560$  280 (syst).  $\Delta Q(\beta^-) = 450$ ,  $\Delta S(n) = 360$  (syst, 2021Wa16).  
 $S(2p) = 417$  105,  $Q(\varepsilon) = 15950$  105,  $Q(\varepsilon p) = 15866$  105, from mass excesses of 4777 105 for  $^{35}\text{Ca}$  measured by 2023La09;  $-9384.3$  4 for  $^{33}\text{Ar}$ ,  $-11172.9$  5 for  $^{35}\text{K}$ , and  $-18378.29$  8 for  $^{34}\text{Ar}$  from 2021Wa16. Values from 2021Wa16:  $S(2p) = 00$  200 (syst),  $Q(\varepsilon) = 16360$  200 (syst),  $Q(\varepsilon p) = 16280$  200 (syst).  
 $S(2n) = 41980$  450 (syst) (2021Wa16).  
Isotope discovery (2011Am01):  $^{40}\text{Ca}(^3\text{He}, \alpha n)^{35}\text{Ca}$  at Berkeley (1985Ay01).  
 $^{35}\text{Ca}$  production:  
1986La17, 1986AnZV:  $\text{Ni}(^{40}\text{Ca}, X)$  at GANIL.  
 $^{35}\text{Ca}$  decay measurements:  
1999Tr04, 1998Le45:  $\text{Ni}(^{40}\text{Ca}, X)$  at GANIL. Measured  $T_{1/2}$  and  $\varepsilon + \beta^+$ -delayed 1p/2p emissions.  
 $^{35}\text{Ca}$  mass measurements: 2023La09, 1985Ay01.  
Theoretical calculations: 2003Sm02, 1998Co30, 1997Co19, 1991De26, 1990Br26.

 $^{35}\text{Ca}$  LevelsCross Reference (XREF) Flags

A  $^1\text{H}(^{37}\text{Ca}, t)$   
B  $^9\text{Be}(^{36}\text{Ca}, ^{35}\text{Ca})$   
C  $^9\text{Be}(^{37}\text{Ca}, X)$

| E(level)              | $J^\pi$ | $T_{1/2}$ | XREF | Comments                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |
|-----------------------|---------|-----------|------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 0.0                   | $1/2^+$ | 25.7 ms 2 | AB   | $\% \varepsilon + \% \beta^+ = 100$ ; $\% \varepsilon p = 95.8$ 3; $\% \varepsilon 2p = 4.2$ 3<br>$J^\pi$ : $L(^{36}\text{Ca}, ^{35}\text{Ca}) = 0$ from $0^+$ .<br>$T_{1/2}$ : From implant-decay correlation in 1999Tr04. Other: 50 ms 30 estimated by comparison with the $^{22}\text{Al}$ yield from 1985Ay01.<br>$\% \varepsilon p$ and $\% \varepsilon 2p$ are derived from the renormalization of $\% \Sigma I(1p) + \% \Sigma I(2p) = 100.6$ in 1999Tr04 to 100. The original decay branching ratios in 1999Tr04:<br>$\% \Sigma I(1p) = 96.4$ 18 and $\% \Sigma I(2p) = 4.2$ 3. |
| $2.09 \times 10^3$ 10 | $3/2^+$ |           | A C  | E(level): weighted average of $2.24 \times 10^3$ 33 from $^1\text{H}(^{37}\text{Ca}, t)$ and $2.08 \times 10^3$ 10 from $(^{37}\text{Ca}, X)$ .<br>$J^\pi$ : $L = 0$ from $3/2^+$ in $^1\text{H}(^{37}\text{Ca}, t)$ .                                                                                                                                                                                                                                                                                                                                                                  |